Avery Matherne

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EDUCATION

Louisiana State University Expected May 2028
M.S., Electrical and Computer Engineering (thesis track), communications and signal processing

Louisiana State University May 2026
B.S., Computer Engineering, minor in Computer Science

- Tiger Excellence Scholars Award, all four years
- Tiger Alumni Legacy Scholarship, all four years
- Coursework: Signals and Systems, Digital Communication and Networking, Probability for ECE, Communications in Computing, Communications Engineering Design

RESEARCH AND ENGINEERING EXPERIENCE

Graduate Research Assistant, Louisiana State University, Electrical and Computer Engineering 2026 to present

- Support LSU's side of Intelligent Data Management for Distributed Naval Platforms, a project building decision-making tools that help distributed autonomous maritime systems decide which data matters and how to share it.

Engineering Intern, Integer Technologies May 2026 to Aug 2026

- Characterized the acoustics of a thruster on a test stand in the lab tank: planned the measurements, ran them, and analyzed the recordings.
- Worked with the underwater network simulation tools.

TECHNICAL PROJECTS

RAMpage Combat Robot, Senior Design Capstone Aug 2024 to May 2025
Lead programmer and systems integration · 1st place overall, Bengal Bot Brawl

- Wrote the control software for a 30 lb combat robot: MicroPython on a Raspberry Pi Pico W and Python on a Raspberry Pi 4B.
- Designed a split architecture: the Pico W handles the weapon from direct RC PWM input, and the Pi 4B streams video on its own.
- Built pneumatic weapon sequencing with a kill switch and signal-loss stops. The firmware allows 17 activations; the tanks give about 6 above 30 lbf.

V2V Collision Warning over OFDM Jan 2026 to May 2026
EE 4003 final project, with lab partner Jason Phan

- Designed and wrote the transmitter, receiver and warning logic in MATLAB: a BPSK-OFDM link modeled on IEEE 802.11p, run over the air on two ADALM-PLUTO SDRs.
- Receiver: preamble detection, timing, CFO correction, a phase fit, a decision-directed PLL, OFDM channel estimation, MMSE equalization and Viterbi decoding.

16×16 LED Audio Spectrum Visualizer 2025
Personal project

- Built a real-time spectrum visualizer on an ESP32 with a 128-point FFT mapped to 16 log-spaced bands.
- Added an adaptive noise baseline and debounced mode switching between the spectrum and an idle glow.

Landspace, Land Research and Mapping App 2025 to present
Personal project

- Building a local-first web and desktop app (React, Leaflet, Electron) for mineral title work.
- Imports messy Excel runsheets, maps parcels, and flags problems in the chain of title automatically.

PROFESSIONAL EXPERIENCE AND LEADERSHIP

Web Designer and Developer, Independent Apr 2026 to present

- Built and run four sites: gracemaealterations.com, plvinc.com, becauseofadam.org and nmdomke.github.io.
- Accessible and fast by default: keyboard support, reduced motion, structured data and optimized images.

Co-Founder, Because of ADAM, nonprofit 2020 to present

- Lead fundraising and community outreach, from cornhole tournaments to water safety booths.
- Designed and built becauseofadam.org, where every statistic links to its source.

Land Agent (Contract), Harold J. Anderson Inc. · Lemoine Landman Services · Petro Land Ventures Inc. 2020 to present

- Research ownership and coordinate permitting for oil, gas and renewable energy projects.
- Build and analyze Excel runsheets to trace title and support project development.

SKILLS

Proficient: Digital communications (BPSK, OFDM, synchronization), Software-defined radio (ADALM-PLUTO), MATLAB, Python, MicroPython, C, Git and GitHub, Linux, ESP32, Raspberry Pi, Embedded wiring and soldering, Public records and title research, Excel runsheets, Client communication, Project management, Fundraising and outreach

Familiar: JavaScript, TypeScript, C++, Astro, React, Next.js, Tailwind CSS, GitHub Actions, Playwright, Cloudflare, Vercel, Supabase, Leaflet, OpenCV, BELLHOP, Web Audio, Accessibility (WCAG), SEO and structured data, Headless CMS setup, Signal processing (FFT, filtering, AGC), Channel coding (convolutional, Viterbi), Equalization (MMSE), HTML and CSS, Performance and Core Web Vitals